

ETL PROCESS AND DATA MART DEVELOPMENT FOR BEVERAGE SALES (2021–2023)

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Course Name: Data Management (DATA 635)

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To: Executive Leadership, Beverages R Us

From: Team 3

Date: April 19, 2026

Subject: Standardization of Beverage Order Data and Development of Analytical Data Mart (2021–2023)

Executive Summary

Beverages R Us should immediately adopt the newly developed beverage data mart as the organization's standard platform for analyzing product performance, regional sales, and Vice President accountability. The completed extract, transform, load, (ETL) process consolidates three years of inconsistent order data into a single, structured dataset that supports accurate reporting and trend analysis. Leadership should require all future order data to follow this standardized format, direct the marketing team to evaluate growth targets, particularly within the Hot Beverages category, and use this dataset to guide strategic decisions around product performance and regional operations. Additionally, the company should consider expanding the model in future phases to include more detailed transaction-level data to support forecasting and customer-level insights.

1. Business Problem, ETL Purpose, and Organizational Value

The organization's historical beverage order data from 2021 through 2023 was stored in separate files with inconsistent formatting and minimal documentation. This created challenges in comparing performance across years, identifying high-performing products, and analyzing sales by region or leadership responsibility.

To address this issue, an ETL process was implemented.

Figure 1. ETL Transformation Logic Used to Standardize Multi-Year Beverage Data

```

25
26  INSERT INTO consolidated_beverage_data (
27      Product, Weight, Volume, Caffeine, Per_Unit_Price, Quantity,
28      Volume_Quantity, Weight_Quantity, Revenue_Quantity,
29      Region, State, Country, Category,
30      First_Name, Last_Name, Year
31  )
32  SELECT DISTINCT
33      t.Product,
34      t.Weight,
35      t.Volume,
36      NULL AS Caffeine,
37      t.Per_Unit_Price,
38      t.Quantity,
39
40      (t.Volume * t.Quantity) AS Volume_Quantity,
41      (t.Weight * t.Quantity) AS Weight_Quantity,
42      (t.Per_Unit_Price * t.Quantity) AS Revenue_Quantity,
43
44      t.Region,
45      t.State,
46      t.Country,
47      c.Category_Name,
48      o.First_Name,
49      o.Last_Name,
50      2021 AS Year
51  FROM beverage_orders_2021 t
52  JOIN Category c
53      ON t.Product = c.Product_Name
54  JOIN org_chart_table o
55      ON c.Category_Name = o.Category_Name
56  WHERE o.First_Name IN ('Remi', 'Buford', 'Bodhi', 'Rowan');
57

```

As shown in Figure 1, the ETL process standardizes data across years and calculates key performance metrics used for analysis.

From a business perspective, this process involved:

- Extracting data from three separate yearly files
- Transforming the data into a consistent structure, including standardized fields and calculated performance metrics
- Loading the data into a consolidated dataset for analysis

During the transformation phase, key business metrics were calculated, including:

- Total volume sold ($\text{Volume} \times \text{Quantity}$)
- Total weight sold ($\text{Weight} \times \text{Quantity}$)
- Total revenue ($\text{Per Unit Price} \times \text{Quantity}$)

Additionally, the data was filtered to include only transactions associated with Vice Presidents Remi Olson, Buford Jackson, Bodhi Perry, and Rowan Walsh, ensuring alignment with leadership-level analysis.

The result is a unified dataset (consolidated_beverage_data) and a summarized output file (GX_output_final.csv) that provide consistent, reliable insight into sales performance.

Figure 2. Consolidated Beverage Data After ETL Transformation

Last_Name	First_Name	Year	Category	Product	Country	Region	State	Weight	Volume	Caffeine	Per_Unit_Price	Quantity_Sum	Revenue_Quantity_Sum
Jackson	Buford	2021	Maple Drinks	Maple Mash UP	USA	Northeast	Maine	794	502	NULL	1098.00	72	79056.00
Jackson	Buford	2021	Maple Drinks	Maple Mash UP	USA	Northeast	Massachusetts	794	502	NULL	1098.00	234	256932.00
Jackson	Buford	2021	Maple Drinks	Maple Mash UP	USA	Northeast	New Hampshire	794	502	NULL	1098.00	150	164700.00
Jackson	Buford	2021	Maple Drinks	Maple Mash UP	USA	Northeast	Rhode Island	794	502	NULL	1098.00	164	180072.00
Jackson	Buford	2021	Maple Drinks	Maple Mash UP	USA	Northeast	Vermont	794	502	NULL	1098.00	158	173484.00
Jackson	Buford	2021	Maple Drinks	Sapped	USA	Northeast	Massachusetts	632	374	NULL	932.00	66	61512.00
Jackson	Buford	2021	Maple Drinks	Sapped	USA	Northeast	New Hampshire	632	374	NULL	932.00	66	61512.00
Jackson	Buford	2021	Maple Drinks	Sapped	USA	Northeast	Rhode Island	632	374	NULL	932.00	92	85744.00
Jackson	Buford	2021	Maple Drinks	Sapped	USA	Northeast	Vermont	632	374	NULL	932.00	74	68968.00
Jackson	Buford	2021	Mocktails	Enchanted Euphoria	USA	Northeast	Vermont	303	284	NULL	364.00	146	53144.00
Jackson	Buford	2021	Mocktails	Enchanted Euphoria	USA	South	Kentucky	303	284	NULL	364.00	106	38584.00
Jackson	Buford	2021	Mocktails	Enchanted Euphoria	USA	South	Louisiana	303	284	NULL	364.00	266	96824.00
Jackson	Buford	2021	Mocktails	Enchanted Euphoria	USA	West	Oregon	303	284	NULL	364.00	146	53144.00
Jackson	Buford	2021	Mocktails	Secret Synergy	USA	South	Mississippi	312	479	NULL	523.00	102	53346.00
Jackson	Buford	2021	Mocktails	Secret Synergy	USA	West	California	312	479	NULL	523.00	44	23012.00
Jackson	Buford	2021	Soft Drinks	Dreamy Droplets	USA	Midwest	Illinois	330	356	NULL	347.00	90	31230.00
Jackson	Buford	2021	Soft Drinks	Dreamy Droplets	USA	Midwest	Nebraska	330	356	NULL	347.00	96	33312.00
Jackson	Buford	2021	Soft Drinks	Dreamy Droplets	USA	South	Florida	330	356	NULL	347.00	46	15962.00
Jackson	Buford	2021	Soft Drinks	Dreamy Droplets	USA	South	Kentucky	330	356	NULL	347.00	76	26372.00
Jackson	Buford	2021	Soft Drinks	Dreamy Droplets	USA	West	California	330	356	NULL	347.00	28	9716.00
Jackson	Buford	2021	Soft Drinks	Sparkle Splash	USA	Midwest	Wisconsin	287	482	NULL	311.00	182	56602.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	South	Arkansas	303	284	83	364.00	102	37128.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	South	Kentucky	303	284	83	364.00	4	1456.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	South	Louisiana	303	284	83	364.00	154	56056.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	South	Maryland	303	284	83	364.00	196	71344.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	South	North Carolina	303	284	83	364.00	168	61152.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	South	South Carolina	303	284	83	364.00	242	88088.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	West	Washington	303	284	83	364.00	168	61152.00
Jackson	Buford	2022	Mocktails	Secret Synergy	USA	Midwest	Iowa	312	479	40	523.00	66	34516.00
Jackson	Buford	2022	Mocktails	Secret Synergy	USA	Midwest	Kansas	312	479	40	523.00	184	96232.00
Jackson	Buford	2022	Mocktails	Secret Synergy	USA	Midwest	Missouri	312	479	40	523.00	96	50208.00

Figure 2 illustrates the standardized interim dataset used as the foundation for final aggregation and reporting.

This ETL effort delivers several business benefits:

- Improved consistency across multiple years of data
- Clear visibility into product, category, and regional performance
- Faster analysis without manual data reconciliation
- Better decision-making based on standardized metrics

2. Evaluation of Data Granularity

The final data mart is structured at the level of:

- Vice President
- Year
- Category
- Product
- Region and State

with aggregated measures such as total quantity and total revenue.

This level of granularity is appropriate for the organization's current needs because it allows leadership to:

- Compare performance across years (2021–2023)
- Evaluate product-level success within categories
- Analyze regional trends
- Assess Vice President performance

At the same time, the dataset removes individual transaction-level detail, which limits deeper analysis such as customer behavior or order-level trends.

Recommendation:

The current level of granularity should be maintained because it effectively supports strategic reporting and aligns with the purpose of the data mart.

However, for future enhancements, the organization should consider capturing:

- Order or transaction identifiers
- Customer-level data
- More detailed time-based data (e.g., monthly or daily trends)

This would allow for more advanced analytics without disrupting the current reporting structure.

Reducing granularity further (for example, removing products or state-level detail) would weaken the organization's ability to identify performance drivers and is not recommended.

3. Ability to Support Marketing Growth Analysis

The marketing team's objective is to determine whether:

- The Hot Beverages category achieves at least 10% year-over-year growth
- Western region's performance remains stable

The final dataset (GX_output_final.csv) fully supports this analysis.

An analyst can:

- Filter by Category = "Hot Beverages"
- Compare total quantity or revenue across 2021, 2022, and 2023
- Calculate percentage growth between years

- Filter by Region = “West” to evaluate stability

Figure 3. Final Aggregated Data Supporting Year-Over-Year and Regional Analysis

Last_Name	First_Name	Year	Category	Product	Country	Region	State	Weight	Volume	Caffeine	Per_Unit_Pric	Quantity_Sur	Revenue_Quantity_Sum
Jackson	Buford	2021	Maple Drinks	Maple Mash i	USA	Northeast	Maine	794	502	NULL	1098	72	79056
Jackson	Buford	2021	Maple Drinks	Maple Mash i	USA	Northeast	Massachuset	794	502	NULL	1098	234	256932
Jackson	Buford	2021	Maple Drinks	Maple Mash i	USA	Northeast	New Hampsh	794	502	NULL	1098	150	164700
Jackson	Buford	2021	Maple Drinks	Maple Mash i	USA	Northeast	Rhode Island	794	502	NULL	1098	164	180072
Jackson	Buford	2021	Maple Drinks	Maple Mash i	USA	Northeast	Vermont	794	502	NULL	1098	158	173484
Jackson	Buford	2021	Maple Drinks	Sapped	USA	Northeast	Massachuset	632	374	NULL	932	66	61512
Jackson	Buford	2021	Maple Drinks	Sapped	USA	Northeast	New Hampsh	632	374	NULL	932	66	61512
Jackson	Buford	2021	Maple Drinks	Sapped	USA	Northeast	Rhode Island	632	374	NULL	932	92	85744
Jackson	Buford	2021	Maple Drinks	Sapped	USA	Northeast	Vermont	632	374	NULL	932	74	68968
Jackson	Buford	2021	Mocktails	Enchanted Ei	USA	Northeast	Vermont	303	284	NULL	364	146	53144
Jackson	Buford	2021	Mocktails	Enchanted Ei	USA	South	Kentucky	303	284	NULL	364	106	38584
Jackson	Buford	2021	Mocktails	Enchanted Ei	USA	South	Louisiana	303	284	NULL	364	266	96824
Jackson	Buford	2021	Mocktails	Enchanted Ei	USA	West	Oregon	303	284	NULL	364	146	53144
Jackson	Buford	2021	Mocktails	Secret Syner	USA	South	Mississippi	312	479	NULL	523	102	53346
Jackson	Buford	2021	Mocktails	Secret Syner	USA	West	California	312	479	NULL	523	44	23012
Jackson	Buford	2021	Soft Drinks	Dreamy Dro	USA	Midwest	Illinois	330	356	NULL	347	90	31230
Jackson	Buford	2021	Soft Drinks	Dreamy Dro	USA	Midwest	Nebraska	330	356	NULL	347	96	33312
Jackson	Buford	2021	Soft Drinks	Dreamy Dro	USA	South	Florida	330	356	NULL	347	46	15962
Jackson	Buford	2021	Soft Drinks	Dreamy Dro	USA	South	Kentucky	330	356	NULL	347	76	26372
Jackson	Buford	2021	Soft Drinks	Dreamy Dro	USA	West	California	330	356	NULL	347	28	9716
Jackson	Buford	2021	Soft Drinks	Sparkle Splat	USA	Midwest	Wisconsin	267	492	NULL	311	182	56602
Jackson	Buford	2022	Mocktails	Enchanted Ei	USA	South	Arkansas	303	284	83	364	102	37128
Jackson	Buford	2022	Mocktails	Enchanted Ei	USA	South	Kentucky	303	284	83	364	4	1456
Jackson	Buford	2022	Mocktails	Enchanted Ei	USA	South	Louisiana	303	284	83	364	154	56056
Jackson	Buford	2022	Mocktails	Enchanted Ei	USA	South	Maryland	303	284	83	364	196	71344
Jackson	Buford	2022	Mocktails	Enchanted Ei	USA	South	North Caroli	303	284	83	364	168	61152
Jackson	Buford	2022	Mocktails	Enchanted Ei	USA	South	South Caroli	303	284	83	364	242	88088
Jackson	Buford	2022	Mocktails	Enchanted Ei	USA	West	Washington	303	284	83	364	168	61152
Jackson	Buford	2022	Mocktails	Secret Syner	USA	Midwest	Iowa	312	479	40	523	66	34518
Jackson	Buford	2022	Mocktails	Secret Syner	USA	Midwest	Kansas	312	479	40	523	184	96232
Jackson	Buford	2022	Mocktails	Secret Syner	USA	Midwest	Missouri	312	479	40	523	96	50208
Jackson	Buford	2022	Mocktails	Secret Syner	USA	Northeast	Vermont	312	479	40	523	124	64852
Jackson	Buford	2022	Mocktails	Secret Syner	USA	South	Alabama	312	479	40	523	186	97278
Jackson	Buford	2022	Mocktails	Secret Syner	USA	South	Arkansas	312	479	40	523	22	11506
Jackson	Buford	2022	Mocktails	Secret Syner	USA	South	Louisiana	312	479	40	523	54	28242
Jackson	Buford	2022	Mocktails	Secret Syner	USA	South	North Caroli	312	479	40	523	170	88910

As shown in Figure 3, the aggregated structure allows direct comparison of revenue and quantity across years and regions.

Why the dataset supports this effectively:

- The Year field enables clear time-based comparisons
- Metrics such as Quantity_Sum and Revenue_Quantity_Sum are consistently calculated
- Categories and regions are standardized across all years
- Data is already aggregated, reducing complexity for analysis

In conclusion, yes, Paul can easily use this dataset to answer the marketing question without needing additional restructuring. The dataset is well designed for this type of analysis.

4. Evaluation of Existing vs. Proposed Data Layout

Two possible data structures were considered for the data mart.

Existing Layout:

- Each row represents one observation for a specific year

- Year is stored as a column
- Metrics (quantity and revenue) are consistent across rows

Proposed Layout:

- Each year's data is stored in separate columns
- Metrics are split into multiple year-specific fields

Analysis:

While the proposed layout may simplify quick comparisons across years, it introduces several limitations:

- It is not scalable, as adding new years requires restructuring the dataset
- It reduces flexibility when analyzing data across different dimensions (e.g., product, region)
- It complicates filtering and aggregation
- It does not align with standard data modeling practices

The existing layout, on the other hand:

- Supports flexible querying across all dimensions
- Easily accommodates additional years of data
- Aligns with best practices in relational and dimensional modeling

Recommendation:

The organization should adopt the existing layout.

While the proposed layout may appear convenient for specific comparisons, it limits long-term usability. The existing structure provides greater flexibility, scalability, and analytical power.

Figure 4. Sorted Output Demonstrating Required Reporting Structure

```

SELECT
  Last_Name,
  First_Name,
  Year,
  Category,
  Product,
  Country,
  Region,
  State,
  Weight,
  Volume,
  Caffeine,
  Per_Unit_Price,
  SUM(Quantity) AS Quantity_Sum,
  SUM(Revenue_Quantity) AS Revenue_Quantity_Sum
FROM consolidated_beverage_data
GROUP BY
  Last_Name,
  First_Name,
  Year,
  Category,
  Product,
  Country,
  Region,
  State,
  Weight,
  Volume,
  Caffeine,
  Per_Unit_Price
ORDER BY
  Last_Name,
  First_Name,
  Year,
  Category,
  Product,
  Country,
  Region,
  State,
  Weight,
  Volume,
  Caffeine,
  Per_Unit_Price

```

Last_Name	First_Name	Year	Category	Product	Country	Region	State	Weight	Volume	Caffeine	Per_Unit_Price	Quantity_Sum	Revenue_Quantity_Sum
Jackson	Buford	2021	Maple Drinks	Maple Mash UP	USA	Northeast	New Hampshire	794	502	NULL	1098.00	150	164700.00
Jackson	Buford	2021	Maple Drinks	Maple Mash UP	USA	Northeast	Rhode Island	794	502	NULL	1098.00	164	180072.00
Jackson	Buford	2021	Maple Drinks	Maple Mash UP	USA	Northeast	Vermont	794	502	NULL	1098.00	158	173484.00
Jackson	Buford	2021	Maple Drinks	Sapped	USA	Northeast	Massachusetts	632	374	NULL	932.00	66	61512.00
Jackson	Buford	2021	Maple Drinks	Sapped	USA	Northeast	New Hampshire	632	374	NULL	932.00	66	61512.00
Jackson	Buford	2021	Maple Drinks	Sapped	USA	Northeast	Rhode Island	632	374	NULL	932.00	92	85744.00
Jackson	Buford	2021	Maple Drinks	Sapped	USA	Northeast	Vermont	632	374	NULL	932.00	74	68968.00
Jackson	Buford	2021	Mocktails	Enchanted Euphoria	USA	Northeast	Vermont	303	284	NULL	364.00	146	53144.00
Jackson	Buford	2021	Mocktails	Enchanted Euphoria	USA	South	Kentucky	303	284	NULL	364.00	106	38584.00
Jackson	Buford	2021	Mocktails	Enchanted Euphoria	USA	South	Louisiana	303	284	NULL	364.00	266	96824.00
Jackson	Buford	2021	Mocktails	Enchanted Euphoria	USA	West	Oregon	303	284	NULL	364.00	146	53144.00
Jackson	Buford	2021	Mocktails	Secret Synergy	USA	South	Mississippi	312	479	NULL	523.00	102	53346.00
Jackson	Buford	2021	Mocktails	Secret Synergy	USA	West	California	312	479	NULL	523.00	44	23012.00
Jackson	Buford	2021	Soft Drinks	Dreamy Droplets	USA	Midwest	Illinois	330	356	NULL	347.00	90	31230.00
Jackson	Buford	2021	Soft Drinks	Dreamy Droplets	USA	Midwest	Nebraska	330	356	NULL	347.00	96	33312.00
Jackson	Buford	2021	Soft Drinks	Dreamy Droplets	USA	South	Florida	330	356	NULL	347.00	46	15962.00
Jackson	Buford	2021	Soft Drinks	Dreamy Droplets	USA	South	Kentucky	330	356	NULL	347.00	76	26372.00
Jackson	Buford	2021	Soft Drinks	Dreamy Droplets	USA	West	California	330	356	NULL	347.00	28	9716.00
Jackson	Buford	2021	Soft Drinks	Sparkle Splash	USA	Midwest	Wisconsin	267	492	NULL	311.00	182	56602.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	South	Arkansas	303	284	83	364.00	102	37128.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	South	Kentucky	303	284	83	364.00	4	1456.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	South	Louisiana	303	284	83	364.00	154	56056.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	South	Maryland	303	284	83	364.00	196	71344.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	South	North Carolina	303	284	83	364.00	168	61152.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	South	South Carolina	303	284	83	364.00	242	88088.00
Jackson	Buford	2022	Mocktails	Enchanted Euphoria	USA	West	Washington	303	284	83	364.00	168	61152.00
Jackson	Buford	2022	Mocktails	Secret Synergy	USA	Midwest	Iowa	312	479	40	523.00	66	34518.00
Jackson	Buford	2022	Mocktails	Secret Synergy	USA	Midwest	Kansas	312	479	40	523.00	184	96232.00
Jackson	Buford	2022	Mocktails	Secret Synergy	USA	Midwest	Missouri	312	479	40	523.00	96	50208.00
Jackson	Buford	2022	Mocktails	Secret Synergy	USA	Northeast	Vermont	312	479	40	523.00	124	64852.00
Jackson	Buford	2022	Mocktails	Secret Synergy	USA	South	Alabama	312	479	40	523.00	186	97278.00

Figure 4 demonstrates the structured sorting applied to support consistent and meaningful reporting.

5. Tidy Data and Its Application

Tidy data is a structured format in which:

- Each variable is stored in its own column
- Each observation is stored in its own row
- Each dataset represents a consistent type of information

The current data mart follows tidy data principles because:

- Each row represents a single aggregated observation (by VP, product, region, and year)
- Each column represents a clearly defined variable

The proposed layout does not follow tidy data principles because:

- It spreads a single variable (Year) across multiple columns
- It makes the data more difficult to analyze and manipulate

Maintaining tidy data is important because it ensures:

- Efficient querying using SQL
- Compatibility with analytics tools
- Easier data transformation and reporting

The existing layout aligns with tidy data principles and should be maintained as the standard format.

Conclusion

The ETL process successfully transformed inconsistent, multi-year beverage order data into a unified and structured data mart. This dataset enables accurate analysis of product performance, regional trends, and leadership accountability.

By adopting this data mart as the standard reporting tool, Beverages R Us will improve its ability to make data-driven decisions, evaluate growth initiatives, and maintain consistency across future data collection efforts.

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Technical Appendix

1. Entity Relationship Diagram (ERD)

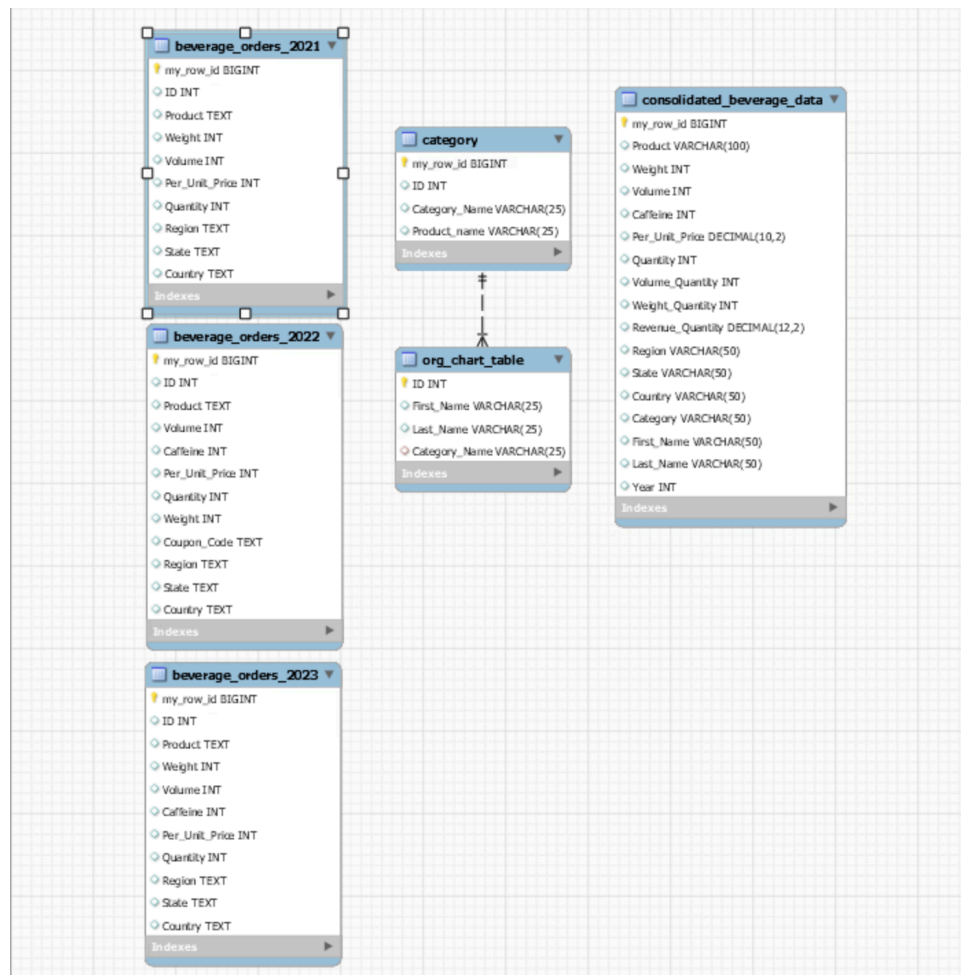


Figure 5. An Entity Relationship Diagram representing the database structure supporting the consolidated beverage data mart.

2. Structure and Purpose of the ERD

The ERD represents the final database design after executing the ETL process, showing how raw beverage order data is transformed, consolidated, and enriched into an analysis-ready dataset. The structure follows a layered architecture that separates raw data, processed data, and supporting reference data.

2.1 Structural Overview

The database is organized into three main components:

A. Source Data Layer

The tables:

- beverage_orders_2021
- beverage_orders_2022
- beverage_orders_2023

These tables store transactional data for each year, including product details, pricing, quantities, and geographic information. While structurally similar, some variations exist across years, requiring standardization during transformation.

B. Consolidation / Staging Layer

- consolidated_beverage_data

This layer represents the transformation step in the ETL process, where data from the yearly tables (beverage_orders_2021, 2022, 2023) is standardized and prepared before being loaded into the final table.

During this step:

- Fields are aligned across all years
- Missing values (e.g., Caffeine in 2021) are handled
- Calculated metrics are created:
 - Volume_Quantity
 - Weight_Quantity
 - Revenue_Quantity
- Data is filtered to include only transactions for Vice Presidents:
 - Remi Olson
 - Buford Jackson
 - Bodhi Perry
 - Rowan Walsh

Unlike a traditional staging layer, this transformation is performed directly within the SQL logic and does not persist as a separate table in the ERD.

C. Final Analytical Output (Data Mart File)

- G3_output_final.csv (*generated from consolidated_beverage_data logic*)

This represents the final analytical output of the ETL process. Although shown in the ERD as consolidated_beverage_data, it functions as a generated file rather than a persistent table.

It includes:

- Vice President attribution (First_Name, Last_Name) from org_chart_table
- The Year field for time-based analysis
- Product category from the category table
- Standardized data from all three years
- Calculated metrics (Quantity_Sum, Revenue_Quantity_Sum)

This dataset is structured to support analysis by:

- Vice President
- Year
- Category
- Product
- Region and State

Because all transformations, joins, and filters are applied before output, this file is fully enriched and business-ready, enabling reporting such as:

- Year-over-year growth analysis
- Regional performance evaluation
- Leadership accountability

D. Supporting Dimension Tables

Two other tables provide additional context:

- category
Maps each product to a corresponding category (e.g., Hot Beverages), enabling grouping and comparison at the category level.
- org_chart_table
Contains organizational details, linking categories to Vice Presidents through First_Name and Last_Name, which supports leadership-level performance analysis.

These tables are joined with the consolidated data in the final analytical layer to add business context without duplicating data, following standard dimensional modeling practices.

2.2 Relationships and Data Flow

The relationships in the ERD reflect a structured data pipeline:

- Source tables are combined into the consolidated table using SQL union operations
- The consolidated table is enriched by joining:
 - category on product name
 - org_chart_table on category name
- The final table integrates all attributes into a single, analysis-ready dataset

2.3 Purpose of the ERD

The ERD serves a technical purpose:

- Defines how data is organized and stored
- Shows how multiple datasets are integrated into a single model
- Ensures consistency between raw data, transformed data, and final outputs
- Provides a blueprint for scalability and future data integration

The ERD confirms that the system follows best practices in data modeling by:

- Maintaining a normalized structure in intermediate stages
- Supporting a tidy, analysis-ready format in the final data mart
- Allowing additional years of data to be added without redesign

3. SQL Annotations and Pseudocode

Import data using the MySQL Workbench table import wizard, load the following files into their respective tables:

- Beverage_orders_2021.csv → beverage-orders-2021
- Beverage_orders_2022.csv → beverage-orders-2022
- Beverage_orders_2023.csv → beverage-orders-2023

Here is the pseudo code used for this project:

BEGIN ETL PROCESS

STEP 1: PREPARE TABLES

Delete consolidated_beverage_data if it already exists
Rename beverage-orders-2021 to beverage_orders_2021
Rename beverage-orders-2022 to beverage_orders_2022
Rename beverage-orders-2023 to beverage_orders_2023

STEP 2: CREATE CONSOLIDATED DATA TABLE

Create consolidated_beverage_data with fields for:

Product
Weight
Volume
Caffeine
Per_Unit_Price
Quantity
Volume_Quantity
Weight_Quantity
Revenue_Quantity
Region
State
Country
Category
First_Name
Last_Name
Year

STEP 3: LOAD AND TRANSFORM 2021 DATA

Read records from beverage_orders_2021

Set Caffeine to NULL

Calculate:

$\text{Volume_Quantity} = \text{Volume} \times \text{Quantity}$

$\text{Weight_Quantity} = \text{Weight} \times \text{Quantity}$

$\text{Revenue_Quantity} = \text{Per_Unit_Price} \times \text{Quantity}$

Join Product to Category table

Join Category to org_chart_table

Keep only records for Vice Presidents:

Remi

Buford

Bodhi

Rowan

Label records with Year = 2021

Insert distinct rows into consolidated_beverage_data

STEP 4: LOAD AND TRANSFORM 2022 DATA

Read records from beverage_orders_2022

Keep Caffeine from the source data

Calculate:

$\text{Volume_Quantity} = \text{Volume} \times \text{Quantity}$

$\text{Weight_Quantity} = \text{Weight} \times \text{Quantity}$

$\text{Revenue_Quantity} = \text{Per_Unit_Price} \times \text{Quantity}$

Join Product to Category table

Join Category to org_chart_table

Keep only records for Vice Presidents:

Remi

Buford

Bodhi

Rowan

Label records with Year = 2022

Insert distinct rows into consolidated_beverage_data

STEP 5: LOAD AND TRANSFORM 2023 DATA

Read records from beverage_orders_2023

Keep Caffeine from the source data

Calculate:

$\text{Volume_Quantity} = \text{Volume} \times \text{Quantity}$

$\text{Weight_Quantity} = \text{Weight} \times \text{Quantity}$

$\text{Revenue_Quantity} = \text{Per_Unit_Price} \times \text{Quantity}$

Join Product to Category table

Join Category to org_chart_table

Keep only records for Vice Presidents:

Remi

Buford

Bodhi

Rowan

Label records with Year = 2023

Insert distinct rows into consolidated_beverage_data

STEP 6: REVIEW CONSOLIDATED DATA

Display all rows from consolidated_beverage_data

Sort rows by:

Last_Name ascending

Year ascending

Revenue_Quantity descending

STEP 7: CREATE FINAL SUMMARIZED OUTPUT

Access data from the consolidated_beverage_data table

Group records by:

Last_Name (Vice President)

First_Name (Vice President)

Year

Category

Product

Country

Region

State

Weight

Volume

Caffeine

Per_Unit_Price

For each group:

Quantity_Sum = sum of Quantity

Revenue_Quantity_Sum = sum of Revenue_Quantity

Output the following columns:

Last_Name

First_Name

Year

Category

Product

Country

Region

State

Weight

Volume

Caffeine

Per_Unit_Price

Quantity_Sum

Revenue_Quantity_Sum

Sort output by:

Last_Name (ascending)

First_Name (ascending)

Year (ascending)

Category (ascending)

Product (ascending)
Country (ascending)
Region (ascending)
Revenue_Quantity_Sum(descending)

STEP 9: EXPORT FINAL OUTPUT

Export results to the final CSV output file

If direct SQL export is restricted, export manually from MySQL Workbench

END ETL PROCESS

After running the final query, export the results to the G3_output_final.csv file using Export Wizard.